Geometry Name______________________
Unit 2 Date______________________
Lesson 8 Homework Period_____________________

Complete the two-column proof.

Given: $\overline{AB} \cong \overline{CD}$

Prove: $\overline{AC} \cong \overline{BD}$

	Statement	Reason
1.	$\overline{AB} \cong \overline{CD}$	Given
2.		Reflexive Property
3.		
4.	$AB + BC = AC$ $CD + BC = BD$	
5.	$CD + BC = AC$	Substitution Property
6.		Substitution Property
7.	$\overline{AC} \cong \overline{BD}$	

A partially completed two-column proof is shown. Match each statement on the left with the correct reason to complete the two-column proof by using reasons provided in the bank. Some reasons may be used more than once.

Given: W is the midpoint of $\overline{VX}$, Z is the midpoint of $\overline{RP}$, and $\overline{VX} \cong \overline{RP}$

Prove: $\overline{WX} \cong \overline{ZP}$

<table>
<tr><th colspan="12">Reasons</th></tr>
<tr><td colspan="3">Given</td><td colspan="3">Combine like terms</td><td colspan="3">Definition of congruence</td><td colspan="3">Definition of midpoint</td></tr>
<tr><td colspan="4">Division property of equality</td><td colspan="4">Segment addition postulate</td><td colspan="4">Substitution property of equality</td></tr>
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	Statement	Reason
1.	W is the midpoint of $\overline{VX}$, Z is the midpoint of $\overline{RP}$, and $\overline{VX} \cong \overline{RP}$	
2.	$\overline{VW} \cong \overline{WX}$ $\overline{RZ} \cong \overline{ZP}$	
3.	$VX = RP$	
4.	$VW + WX = VX$ $RZ + ZP = RP$	
5.	$VW = WX$ $RZ = ZP$	
6.	$VW + WX = RZ + ZP$	
7.	$WX + WX = ZP + ZP$	
8.	$2WX = 2ZP$	
9.	$WX = ZP$	
10.	$\overline{WX} \cong \overline{ZP}$	